Using Natural Language Inference to Improve Persona Extraction from Dialogue in a New Setting

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Abstract

Developing persona-grounded dialogue agents requires training models to converse conditioned on a provided persona. Hand-crafting these persona is time-consuming and expensive, so researchers leverage *persona extraction* to automatically extract persona information from past character dialogue. Existing labeled persona extraction data is set in "real world" chit chat, and models trained on this data are not well-suited for other narrative settings like fantasy worlds. We introduce two natural language inference-based (NLI) post-hoc methods to adapt a trained generative persona extraction model to a new narrative setting: (1) generate multiple candidates and re-rank based on dialogue-candidate inference score (NLI Re-rank) and (2) aggressively remove candidates that cannot be inferred from the dialogue (Neutral Removed). Compared to existing persona extraction models, our method returns higher-quality extracted persona and requires less human annotation.

1 Introduction

Persona-grounded dialogue agents are often assigned a persona or a description of a character identity to impersonate while responding to a user (Zhang et al., 2018). These hand-crafted persona descriptions are supplied to the model's context along with the conversation history (Zhang et al., 2018; Urbanek et al., 2019) or via character-specific embeddings added to the encoder (Li et al., 2016b) during response generation. See Figure 1 for examples of character personas.

Constructing persona for these agents can be time-consuming, but *persona extraction* alleviates this by automatically extracting information about a character given past dialogue utterances (Wang et al., 2022; Lu et al., 2022; Zhou et al., 2023; Zhu et al., 2023b). This task is a precursor to developing persona-grounded response generation models, or models whose outputs are consistent with their assigned personas. Manually crafting persona can be arduous, and what a character says can be rich with information. In contrast, current persona extraction models are all trained on the same domain (chit-chat), since the standard training dataset is derived from PersonaChat (Zhang et al., 2018). PersonaChat is considered to exist in the "real world" and thus persona extraction models trained on it have difficulty extracting persona in new and distant narrative settings such as the fantasy world in LIGHT (Urbanek et al., 2019). A failure of domain transfer is shown in Figure 2, where an extraction model trained on chit-chat hallucinates on a fantasy dataset and incorrectly associates a royal character "prince" with the musician Prince.

A model that generalizes well to various narratives is desirable due to the high cost of manual annotation for each new narrative setting. We thus exploit post-hoc methods on a trained model to mitigate domain adaptation issues. Our approach

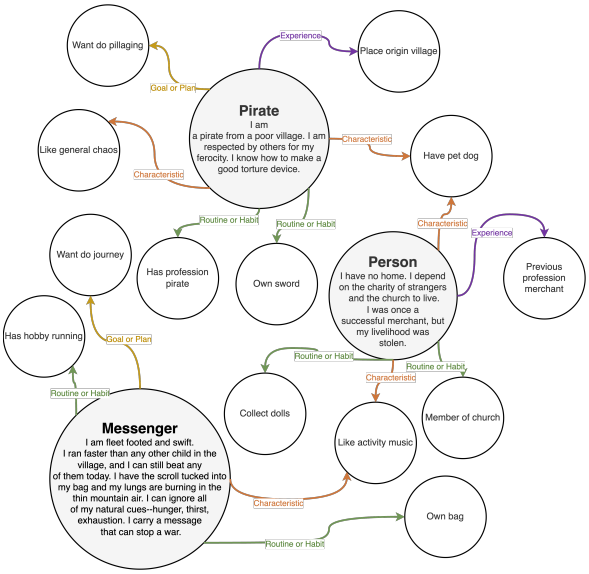


Figure 1: PeaCoK-style persona knowledge graph Gao et al. (2023a) for a few characters from the LIGHT fantasy role-playing dataset. The graph was built from character utterances in conversations.

is to cast it as a natural language inference (NLI; Welleck et al. (2019)) task, as we want to ensure that extracted persona are reasonable given the dialogue utterances.

We start with a new persona extraction model trained on PersonaExt-PeaCoK, a semi-automatic labeled dataset we modified from PersonaExt (Zhu et al., 2023b), of (utterance, persona triplet) pairs. An example datapoint is ("i am going to auburn for med school", (i, routine or habit, attend school auburn)). The utterances were originally derived from PersonaChat (Zhang et al., 2018), which are clearly different from the subject and style of speaking in other settings (e.g., "Blood from a book, what is this magic you speak of?" from fantasy dataset LIGHT (Urbanek et al., 2019), Figure 3). We modified the training dataset for compatibility with PeaCoK (Gao et al., 2023b), a generalizable knowledge-grounded representation for personas. Similar to prior work, we cast the persona extraction task as a sequence-to-sequence (seq2seq) problem, mapping a dialogue utterance to a persona knowledge "triplet" (Ni et al., 2022; Wang et al., 2022; Zhu et al., 2023b) (see Figure 2).

Persona generation from out-of-narrative (i.e., from a narrative setting different from the training data) dialogue utterances is challenging due to model hallucination or generation of low-quality, generic, or incorrect persona information. We thus introduce two post-hoc methods that can be used to improve the quality of persona extracted from dialogue that differs from the original model training data. Our methods use a new persona-centric NLI model to re-rank or remove unreasonable personas given the utterance. The converse relationship between utterances and persona has been investigated before, and even used to make persona extraction datasets (Wang et al., 2022), but the direction we explore allows us to determine if the extracted persona can be inferred from the utterance. We explore two variants: (1) generate many persona candidates and re-rank with the NLI model (**NLI Re-Ranked**); and (2) starting from variant (1), remove all candidates that are not considered entailed by the NLI model (**Neutral Removed**). See Figure 3 for an overview of the process.

We found that the Neutral Removed post-hoc method largely reduces false positives (i.e., hallucinating persona information from an utterance) and improves the quality of extracted persona when compared to the current state-of-the-art model

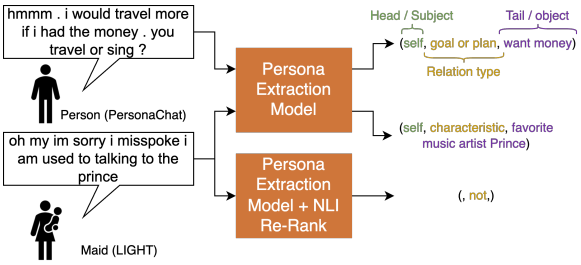


Figure 2: Example domain-transfer failure of a persona extraction model trained on PersonaChat-derived data on fantasy LIGHT dialogue. Our proposed NLI Re-Ranking approach remedies this hallucination.

PAED (Zhu et al., 2023b) and to not using the method.

Our **contributions** are[1]:

- Two post-hoc methods for improving the quality of persona extraction on a new narrative setting: **NLI Re-Ranked** and **NLI Neutral Removed**;
- **PersonaExt-PeaCoK**, a semi-automatically adapted dataset from PersonaExt (Zhu et al., 2023b) for training PeaCoK-compatible persona extraction models

2 Related Work

Our work is relevant in the focus areas of persona-grounded dialogue, persona extraction, and dialogue-specific natural language inference (Dialogue-NLI).

Persona-grounded Dialogue The primary issues with dialogue agents are their boring and generic responses (Li et al., 2016a; Khayrallah and Sedoc, 2021) and their inability to maintain a consistent persona, often contradicting themselves in conversations (Shuster et al., 2022). An early, widely-used work to address this issue is PersonaChat (Zhang et al., 2018) which is a manually created dataset of crowdworker conversations paired with 3-4 sentence persona descriptions. Similarly, Learning in Interactive Games with Humans and Text (LIGHT; Urbanek et al. (2019)) is a dataset created the same way and modeled after PersonaChat, but is set in a fantasy world including characters like fairies, pirates, and kings. While other fantasy dialogue datasets exist, either they contain prose instead of dialogue (Zhu et al., 2023a) and/or do not have structured persona information paired

[1]All code and models will be released upon publication.

with each character (Callison-Burch et al., 2022; Weir et al., 2023; van Stegeren and Theune, 2020). Appendix Table 3 reviews the datasets in detail. A crucial yet overlooked component of persona-grounded datasets is persona *commonsense knowledge* grounding. Gao et al. (2023a) fill this gap with PeaCoK, formalizing relations within and between personas. Personas are formulated as knowledge triplets consisting of head and tail entities connected by a relation, e.g, (I [Pirate], goal or plan, want do pillaging), see Figure 1.

Persona Extraction The task of persona extraction is new, and methods often draw from the established field of relation extraction. Wang et al. (2022) separate persona extraction into extraction and inference subtasks, or if the relation can be found verbatim in the utterance. They trained GPT-2 Small on Dialog-NLI (Welleck et al., 2019) and re-rank generated output based on an external model (not NLI).

Ni et al. (2022) and Zhu et al. (2023b) also cast persona extraction as a seq2seq mapping from dialogue utterance to a persona triplet. Zhu et al. (2023b) used a variational auto-encoder (VAE) to create new synthetic examples to train the model to be able to distinguish between related relations ("like" and "dislike"). They introduced the BART-based zero-shot model PAED, and also released a modified Persona Extraction dataset (PersonaExt) built from the persona extraction dataset from Wang et al. (2022). Related to persona extraction is persona *expansion* (Liu et al., 2022; Kim et al., 2023), where commonsense KGs are leveraged to add more information given a presented persona (typically from PersonaChat). Character identification is also related to persona extraction, where a speaker is identified given their persona information and/or past dialogue (Sang et al., 2022).

In summary, while we also use seq2seq modeling for persona extraction, we introduce NLI-based reranking and filtering for quality control in a new narrative setting (from "real world" to fantasy). Our persona extraction framework is built upon the high quality persona commonsense knowledge graph PeaCoK (Gao et al., 2023a), further ensuring its applicability across various new domains.

Persona NLI The Dialog-NLI dataset (Welleck et al., 2019) is created to facilitate NLI problems specifically in the narrative or dialogue domain. It consists of manually annotated (persona sentence, character dialogue utterance) pairs for their entailment relationship from PersonaChat. We incorporate an NLI model, dubbed Persona-NLI, for accurate persona extraction from dialogue utterances. Since Dialog-NLI is built from PersonaChat, which is also what PersonaExt was built from, we do not include it to train the Persona-NLI model to avoid possible train/test leakages. Instead, we train Persona-NLI on the train split of PersonaExt, augmented with negative examples (Section 4). Ammanabrolu et al. (2021) introduced a commonsense knowledge graph for LIGHT (ATOMIC-LIGHT), but we do not include it since it there is no dialogue and our experimental setting assumes no narrative-specific information.

3 Persona Extraction from Dialogue

Our goal is to automatically extract persona knowledge of characters from their dialogue utterances, where the characters are from an unseen narrative setting. It is challenging to assume the types of relations of a new setting, thus we use existing relation types from PeaCoK (Gao et al., 2023a), the first and only commonsense knowledge graph for personas. As shown in Figure 1, PeaCoK relation types are generally sufficient to fit various narrative setting. PeaCoK itself does not contain dialogue utterances, so we use the PersonaExt (Zhu et al., 2023b) corpus, a dataset of (utterance, persona relation) pairs to train a persona extraction model to output a persona following PeaCoK format.[2] We discuss the adaptation of PersonaExt to PeaCoK persona format in Section 3.1.

We ablated over prompts and structured output templates for the persona extraction model. Our model outperforms the Persona Attribute Extraction in Dialogues (PAED) model (Zhu et al., 2023b) in both the fine-tuned and zero-shot settings. To focus on the main results and analyses of our narrative adaptation methods, we refer the reader to Appendix C for the model details and results on PersonaExt-PeaCoK.

3.1 Dataset: PersonaExt-PeaCoK

PeaCoK is a knowledge graph consisting of head and tail entities with relation types of persona commonsenses, e.g, ("I am a famous pianist", "characteristic", "skilled in playing the piano"), and it does

[2]We use PersonaExt instead of its precursor dataset of Wang et al. (2022) because Zhu et al. (2023b) improved upon the annotations and labels.

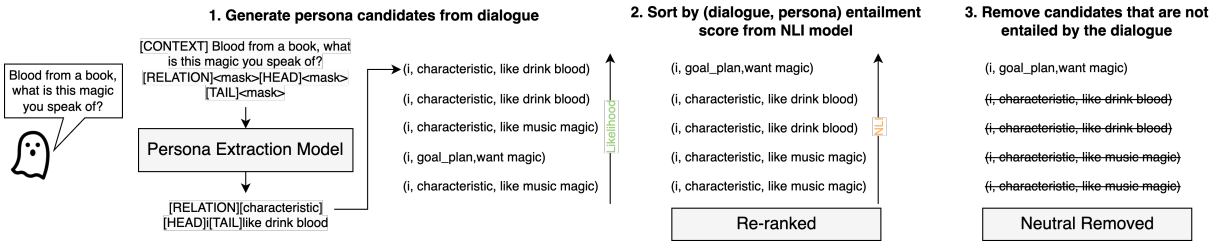


Figure 3: Overview of persona extraction process from dialogue utterances with a trained seq2seq (BART) and NLI model. The head, tail, and relation types are parsed from the model output. As seen from the selected example, the proposed NLI re-ranking step correctly adjusts the order of possible extracted persona, penalizing the non-entailed persona knowledge ("i", "characteristic", "like drink blood").

not provide dialogue utterances. While the heads and tails are phrases, combining them into full sentences would lead to repetitive and unnatural utterances. Instead, we start from an annotated dialogue dataset, PersonaExt, and semi-automatically convert them into PeaCoK format by re-annotating the dialogue utterances on the *relation level* to PeaCoK relations. We refer to the relabeled PersonaExt dataset as **PersonaExt-PeaCoK**. The train, validation, and test sets for model training were stratified-split according to the labels (28412/3157/3508 examples, respectively). See Appendix B.1 for more dataset preparation details.

3.2 Extraction Model Training

We fine-tuned BART-Large from HuggingFace (Lewis et al., 2020; Wolf et al., 2020) on PersonaExt-PeaCoK.

We created a structured input and output template inspired by PAED (Zhu et al., 2023b), shown in Figure 3. All template and label tokens are added to the vocabulary, and were initialized to the average embedding of their respective short text descriptions from (Gao et al., 2023a) (Appendix Table 6).

Since the extracted persona needs to be in the correct format for parsing into a triplet, we impose constraints on the generation to ensure template adherence. These constraints are flexible and can be combined with any decoding method. For the in-narrative setting (PersonaExt-PeaCoK) we use greedy search since there was no improvement with other decoding methods such as beam search.

We trained the model on four 48GB NVIDIA RTX A6000 GPUs for 50 epochs, 128 batch size, AdamW optimizer, and $5e^{-5}$ learning rate. Training time was 1.5 hours.

Evaluation For evaluating persona extraction, we employ the same accuracy metric as Zhu et al. (2023b), where the model is rewarded only if the entire (head, relation, tail) triplet is predicted correctly (shown in our results as "Triplet"). We additionally relax this metric and evaluate the model performance separately on identifying the head, relation, and tail entities from the triplet, similar to Wang et al. (2022).

According to the reference metrics, our model outperforms comparative model PAED (triplet accuracy 0.66 vs 0.61), and we omit comparisons to PAED for the narrative adaptation experiments. See Appendix C.3 for detailed results.

4 NLI for Post-hoc Narrative Adaptation

As shown in Figure 2, a persona extraction model trained on the real-world chit-chat setting will hallucinate when inferring on a different setting. These hallucinations can lead to numerous erroneous persona. This is expected as persona extraction training data does not contain many negative examples (i.e., statement/utterance without a persona) or "filler" utterances (e.g., "Okay"). As a result, it is crucial to have an automatic method for filtering out "hallucinated" persona output from the model.

We leverage natural language inference (NLI) – whether a statement *follows* or can *be inferred from* another statement (MacCartney and Manning, 2008) – for filtering out the hallucinated persona. To alleviate potential issues caused by a domain shift between dialogue and general NLI tasks (Bowman et al., 2015; Williams et al., 2018), we train an NLI model specifically for persona extraction, **Persona-NLI**.

4.1 Persona-NLI Model

To create Persona-NLI, we fine-tuned an existing NLI model on a new persona- and dialogue-centric dataset. We use `nli-deberta-v3-base`

(Reimers and Gurevych, 2019)[3], which was trained on SNLI (Bowman et al., 2015) and MNLI (Williams et al., 2018). While MNLI does contain training data in the fiction domain (≈20%), it is prose from contemporary fiction and not dialogue utterances.

Persona-NLI Dataset We created a modified version of the PersonaExt-PeaCoK dataset designed for NLI training. For each training example the premise was the dialogue utterance and the hypothesis was the ground-truth persona in `HEAD : TAIL : RELATION` format. The dataset consisted mostly of positively entailed examples. To augment the dataset with negative examples and to add robustness to relation types, we added an example for every other relation type. Two example training instances are ("yeah as long as i can do some katy perry", "i : like music katy perry : characteristic", TRUE) and ("yeah as long as i can do some katy perry", "i : like music katy perry : goal_plan", FALSE). This augmentation resulted in train, validation, and test splits of 142060, 15785, and 17540 examples, respectively.

Training and Results Starting from an existing high-performing NLI model, "Base-NLI" (`nli-deberta-v3-base`), we fine-tune on our Persona-NLI dataset for 5 epochs, batch size of 32, learning rate of $2e^{-5}$, and AdamW optimizer on a single 48GB NVIDIA RTX A6000 GPU. Persona-NLI achieved a macro-F1 of 0.60 on the test data, which is a large increase compared to the base model (without fine-tuning) with an F1 of 1.0.

4.2 NLI-Reranking

A benefit of a generative persona extraction model is the ability to sample many candidates. We take advantage of this ability with a *generate-many and re-rank* approach, dubbed **NLI-Reranking**. To sample diverse candidates we use diverse beam search (Vijayakumar et al., 2018), a search algorithm designed for greater exploration of the search space than beam search, leading to greater diversity. We use 5 beams and set the number of beam groups to 5 and diversity strength $\lambda = 0.4$.

As shown in Figure 3, we can adjust the model's final generated output by first generating multiple candidates, scoring each candidate with the Persona-NLI model, and then selecting the one with the new highest score.

[3] https://www.sbert.net/docs/pretrained_cross-encoders.html#nli

The NLI score for each candidate is based on the entailment score of (utterance, candidate triplet) pairs. If the pair is determined to have an entailment relationship, we then adjust the final score of the candidate as the language modeling score (probability of the sequence) plus the probability of entailment. To calculate the probability of each candidate we use softmax over all candidates for the utterance. The candidate probabilities are then added to the probabilities from the Persona-NLI model, and adjusted with softmax again.

If there is no entailment relation, the score is kept the same, i.e., only the language modeling score. This new score is designed to promote candidates that have high entailment scores, but not penalize the candidates determined to be highly likely by the language model.

This method can only be used with a model that can output multiple candidates, such as BART with a beam search or sampling-based decoding methods, and not with greedy search.

4.3 NLI Neutral Removed

A stricter use of the NLI model is to completely remove any candidates that cannot be entailed by the utterance, i.e., has a neutral or contradictory relationship. A "removed" candidate is replaced with "(,not,)" (see Figure 2) which indicates that no relation was extracted from the utterance. The idea for this strict removal is to reduce hallucinations by improving precision at the expense of recall. Since there is no re-ranking, it can be used by models that only generate one candidate. We refer to this as **Neutral Removed**.

5 Evaluation on a New Setting

We combine the trained persona extraction model and NLI post-hoc methods on a narrative setting different from the chit-chat training data: the fantasy world of LIGHT. Since this dataset does not have ground truth like PersonaExt-PeaCoK, we evaluate with intrinsic and human evaluations (Section 5.2).

5.1 Dataset: LIGHT

As mentioned in Section 2, LIGHT is a PersonaChat-esque dataset where workers are asked to embody a persona from a fantasy world for conversations with other workers. Unlike PersonaExt-PeaCoK, this dataset contains many "filler" sentences that do not contain any persona information, e.g., "Hello". To help remedy this,

		Per Character					Overall
		Persona Recall	Cov.	1st Person	Unique Persona	# Persona	Ratio Accepted
Neutral Removed	Diverse Beam	0.25	0.50	**0.88**	0.92	37.66	**0.60**
	Greedy	0.28	0.63	0.87	0.92	47.61	0.55
NLI Re-Rank	Diverse Beam	0.33	**0.88**	0.86	0.92	66.80	0.58
Baseline	Diverse Beam	**0.34**	**0.88**	0.86	0.92	67.09	0.57
	Greedy	**0.34**	0.86	0.86	0.92	65.53	0.56

Table 1: Summary results from the post-hoc NLI method as compared to the Base model on the new narrative setting (fantasy, LIGHT). "Cov." is short for utterance coverage. Inter-annotator agreement (IAA, Krippendorff's Alpha measured with two annotators) of 0.65. "Ratio Accepted" refers to extracted persona annotated with the "Yes" category (i.e., marked as appropriate by both annotators).

we removed dialogue utterances with less than 5 words. Also, to minimize narrative transfer issues we lowercased and tokenized the utterances similar to those in PersonaExt-PeaCoK with NLTK `word_tokenize` (Bird and Loper, 2004).

Due to the large size of the dataset (1377 unique characters and 19334 total dialogue utterances), we analyze LIGHT on a per-character basis instead of on a per-utterance basis. We do this by pairing the 3-4 sentence character persona description and the extracted persona from the dialogue across *all* character conversations. An example character description is in Table 2 (top). This evaluation is directly comparable to the model use case, where extracted persona would be manually verified before being used with a dialogue agent in a production setting. More dataset processing details are in Appendix B.2.

5.2 Evaluation without Ground-Truth

Since there are no human annotated ground-truth persona triplets for LIGHT, we developed intrinsic (i.e., non-reference-based) metrics alongside qualitative analysis.

Intrinsic Metrics The non-reference-based metrics focus on the number of returned persona and unique persona. The metrics are measured for each character, overall, and on the dialogue and description datasets.

We define utterance **coverage** as the ratio of extracted persona to the number of utterances. If the extracted persona (triplet) has the "not a relation" relation type, it is not considered. We are curious about how many extracted relations are about the speaking character or other characters. This **first person** metric measures the ratio of first-person triplets, where the triplet head is "I", "me", or "my", as compared to all extracted triplets.

We look at the rate of unique head and tail entities, referred to as **unique head** and **unique tail**.

Persona "Recall" While there are no ground-truth labels for LIGHT on the utterance-level, we can consider the persona descriptions as "silver" labels. We define **recall** as the ratio of persona relations successfully recovered from the utterances. The persona relations are extracted using without any post-hoc NLI methods, since every sentence in the description contains persona information.

Human Evaluation For the qualitative analysis, one author and one non-expert non-author compared the extracted persona information extracted from the dialogue utterances to the character description itself. Each triplet was annotated against the provided character name and 3-4 sentence description (i.e., persona profile in Section 5.1). The overall categories were "Yes", i.e., accept the triplet for the character, and "No", i.e., reject the triplet. Within the overall categories, subcategories were provided for finer-grained evaluation. Under "Yes", the persona information could be *directly* related to the persona description or *reasonable* given the description. If the triplet is rejected, the annotator must choose between the following options: *contradictory*, *unreasonable*, and *malformed*. For human evaluation, we annotated triplets from 10 randomly sampled characters across all the models (3569 generations with 1200 unique relations). See Table 2 for an example and Appendix D for annotation details.

5.3 Results

Table 1 shows the results, and detailed results are in Appendix E. The overall annotations (i.e., "Yes" or "No") had an inter-annotator agreement (IAA) of 0.65 and an IAA of 0.54 for the detailed annota-

Character Description I silently swoop down into the forest to carry off another. I do what I can to please the high priestess, who is my mother. I feast on the bones of the hunter. I like being scratched behind my ear and laying by the fire.	
Extracted Persona Triple from Dialogue	**Annotation**
(self, characteristic, like food nutritious)	Yes – R
(self, goal_plan, want do hurts)	Yes – R
(self, characteristic, like food venison)	Yes – R
(self, routine_habit, has profession priestess)	No – C
(self, characteristic, like activity play)	Yes – R

Table 2: Example human annotations for extracted persona triplets for the "a dragon" character in LIGHT. "R": Reasonable; "D": Directly; "C": Contradictory. The dialogue was aggregated across multiple conversations for each character.

tions as measured by Krippendorff's Alpha.[4]

NLI Removal has the largest impact. Removing non-entailed triplets reduces the number of extracted triplets (i.e., those not predicted as [no_relation]) by 44% for Diverse Beam search and 27% for Greedy Search when comparing the average number of persona extracted between the Baseline (no post-hoc) and NLI Neutral-Removed methods. This extreme reduction also impacts persona recall and utterance coverage. Persona recall dropped from 0.34 to 0.25 for Diverse Beam search and 0.34 to 0.28 for Greedy Search. Similarly, utterance coverage dropped from 0.88 to 0.50 for Diverse Beam search and 0.86 to 0.63 for Greedy Search.

Most extracted persona are about the character. Across all models there are high ratios of extracted persona that are first-person ($86\%+$), or have head "I", "me", or "my". NLI Neutral-Removed models have the highest number of persona about the given character ($87 - 88\%$), most likely due to the NLI model removing hallucinations about other character names in the dialogue.

Diverse Beam Search with NLI Neutral Removed is the best model. The differences in extracted persona relations are more apparent with human evaluation than from automatic evaluation alone. The number of extracted relations are different for each method, so we focus on the ratio of accepted (i.e., labeled "Yes") relations. With this metric, using Persona-NLI to remove non-entailed persona generated with diverse beam search is the best performing, with 60% of extracted relations being accepted. Interestingly, removing the non-entailed relations with Greedy Decoding hurts performance compared to the Baseline (0.55 accepted versus 0.56 accepted).

[4] https://github.com/LightTag/simpledorff

6 Discussion

From the results we show an example downstream use case (creating a persona knowledge graph, PeaCoK), the effect of "quality vs diversity" phenomena, and a deeper look at the NLI Reranking module.

Character Knowledge Graph One use-case of our persona extraction model is to adapt a knowledge graph (e.g., PeaCoK) given past dialogue from a narrative setting that differs from the original training data. We showcase the ability of our best-performing model by building a graph with a few LIGHT characters Section 5.2, shown in Figure 1. These three characters were selected from the 10 that were annotated for the human evaluation. This graph can then be used for enhancing persona-grounded dialogue as PeaCoK. From the dialogue, the persona extraction model was able to extract persona information beyond the persona. For example, while the Pirate obviously (from the description) works as a pirate and is from a village, they also have a pet dog and own a sword.

Quality vs Diversity Prior work in decoding has shown a "quality vs diversity" trade-off, where after a certain point of trying to increase diversity in generations, the generation quality decreases. In our use case, the diversity from diverse beam search brought higher quality persona than from greedy search alone, even before introducing the NLI post-hoc methods. This indicates that diversity can be beneficial for adapting to a new narrative since it is less biased towards the original training setting.

Another trade-off was "quality over quantity", since the Neutral Removed models returned significantly less persona than the Baseline and Re-ranking models. This was a benefit since there were less personas to annotate (e.g., an average of 37-47 extracted persona per character instead of 65-67 (Table 1)). Since even the best model had an acceptance rate of only 60%, there is still a need for a human evaluation step. This is further enforced by the lack of clear relationship between the intrinsic, quantitative metrics and the manual annotations.

Impact of Persona NLI While we trained an NLI model on the entailment relationship between dialogue utterance and persona, the training domain was still real-world "chit-chat". Thus, the Persona NLI model is also being used in a zero-shot narrative setting alongside the persona extraction model. This indicates that the NLI post-hoc methods with higher performing NLI models could improve the quality of extracted persona.

7 Conclusion

Our goal was to address the challenges in adapting a persona extraction model trained on one narrative setting (e.g., real-world "chit-chat") to another setting (fantasy). We modeled persona extraction as a seq2seq problem and fine-tuned BART on PersonaExt, a dataset of (utterance, persona) pairs built from Dialog NLI and PersonaChat. We converted the "chit-chat" PersonaExt to the general relations from PeaCoK (e.g., "characteristic") to extract persona information applicable to any narrative setting. With our trained persona extraction model, we evaluated two NLI-based post-hoc techniques to adapt the model to dialogue from a different narrative setting, the fantasy world of LIGHT. Our two approaches were (1) NLI Re-Ranking, where we generate many persona candidates and re-rank them based on their entailment scores, and (2) a strict Neutral-Removed method where candidates not entailed from the dialogue utterance are removed from re-ranking. We found that leveraging inference information to remove persona candidates that can not be inferred from the utterance worked the best according to human evaluation (Neutral-Removed).

Limitations

The main limitation of this work is assuming that existing dialogue from a character is available to extract persona information from. Further, our methods are only evaluated on dialogue utterances and not on other character-related text such as prose (see Appendix A).

Ethical Considerations

The ethical concerns of this work center on the possibility of automatically impersonating an existing person, rather than the intended use case of fictional characters. Further, our model is trained on the PersonaExt dataset (derived from the crowdsourced PersonaChat), so we cannot guarantee no presence of offensive language. Manual evaluation is always the best final step, and we encourage developers who use our method for persona extraction to add a toxicity (e.g., hate or offensive speech) evaluation step in addition to quality evaluation.

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ARR Re-Submission Notes

A previous iteration of this work was submitted to ARR February 2024. The following changes have been made for submission to ARR June 2024:

- Re-structuring of the paper to highlight main contributions of NLI post-hoc methods. This re-structuring was mostly accomplished by moving the persona extraction model training information to the Appendix and expanding the NLI adaptation section.
- Removing the Beam Search results since they were very similar to Diverse Beam Search. We now only compare Diverse Beam Search and Greedy Search.
- Training a new Persona-NLI model that is based on an augmented version of the PersonaExt-PeaCoK training dataset instead of ComFact. This NLI model also includes the relation type when predicting entailment.
- Changing the calculation of the final scores of the NLI post-hoc re-ranking method. Instead of adding the logits from the trained BART model to the logits from the NLI model, we now (1) softmax the BART sequence logits (2) softmax the NLI entailment/neutral logits (3) Add the BART probabilities and NLI probabilities and (4) softmax the added scores.
- General writing improvements.
- Consolidation of appendices for improved readability.

All changes were made in response to the ARR February 2024 reviewers. We thank them for their time and helpful comments.

A Narrative-Specific Datasets

While there are many existing datasets to support narrative analyses, we require a dataset that supports our tasks of *persona extraction from dialogue* and *dialogue agent roleplay*. The datasets we considered are in Table 3. The datasets varied in size, type of dialogue, and type of persona description. The dialogue from a character can either be presented in prose or conversational dialogue form, e.g., "as the knight slew the dragon he yelled "for glory!'" versus "Knight: For glory!". Also, to evaluate our extracted person graph we need ground-truth persona data in the form of character-level information. This persona information can either be absent, structured, or unstructured. Structured persona information is either provided as sentences, as in LIGHT (Urbanek et al., 2019), or in tabular format (Li et al., 2020; Zhu et al., 2023a).

B Dataset Details

B.1 PersonaExt-PeaCoK

We modified the PersonaExt dataset (Zhu et al., 2023b) for compatibility with PeaCoK (Gao et al., 2023a) by re-labeling the relations and converting the single-word tails to phrases. We refer to the relabeled PersonaExt dataset as **PersonaExt-PeaCoK**. The train, validation, and test sets for model training were stratified-split according to the new labels (see Table 12 for details.)

Conversion to PeaCoK Labels The 105 PersonaExt labels were mapped to one of four PeaCoK labels: routine or habit, characteristic, goal or plan, and experience. While there is also a "relationship" label, this is a meta-label that can occur between attributes *across* characters, so we do not include it as a label for the extractor. Two authors manually mapped the labels and disagreements were discussed until annotations were unanimous. The relation mappings are shown in Table 11.

Tail Phrase Creation To create the tail phrases we combine a PersonaExt relation with the tail, e.g., "(have_family, wife)" becomes "have family wife." This can create phrases which are not proper English and we leave addressing this to future experiments.

B.2 LIGHT

We download LIGHT from ParlAI (Miller et al., 2017).[5] We found 10,268 dialogues (230 more than Urbanek et al. (2019)) and 1,382 unique characters (e.g., "a baby dragon"). "Unique" characters were determined by counting the unique (character, description) pairs. We ignore the "objects", "actions", "emotes", and "actions" portions of LIGHT since our persona extraction model is based only on dialogue. Persona descriptions are provided as a paragraph of 3-4 sentences, where each sentence contains a character trait. We split the paragraph into individual sentences, and each sentence has one character/persona trait. Dialogue utterances, which may contain more than one sentences, are left as-is.

C Persona Extraction Model Training Details

We fine-tuned BART (Lewis et al., 2020) with the HuggingFace Seq2SeqTrainer (Wolf et al., 2020). The hyperparameter settings are shown in Table 8.

C.1 BART Fine-tuning Templates

We experimented with variations of the structured input and output templates for fine-tuning BART. The special tokens for the templates are added to the model's vocabulary (Table 6). The templates are shown in Table 5 and the results are in Table 7.

Our models perform similarly on the overall accuracy (i.e., triplet accuracy) regardless of the input or output template, other than Relation-first, with scores around 0.55. The model with the Relation-first template had a higher overall accuracy score of 0.60, which is comparable to the score from comparative model PAED (0.61). We use the Relation-first template for all other experiments.

C.2 Comparative Models

We compare our persona extraction model to the state-of-the-art persona extraction model PAED.

PAED (zero-shot) PAED (see Section 2) is also a fine-tuned BART model for persona extraction in zero-shot settings, and we test it with PeaCoK relations. We trained the PAED model using the official codebase with 10 unseen labels and seed 0. The reported results for zero-shot PAED on their PersonaExt dataset was 0.32 for $n = 10$, and our

[5] https://github.com/facebookresearch/ParlAI/tree/main/projects/light

triplet accuracy score of 0.34 is close, despite the tail phrase modification of the dataset.

PAED (fine-tuned) While PAED was designed as a zero-shot model to adapt to new relation types, we also include a fully fine-tuned version that is trained on the same train split of PersonaExt-PeaCoK as our model. We trained the PAED model from scratch on PersonaExt-PeaCoK with the same settings as Zhu et al. (2023b).

C.3 Persona Extraction Model Results

The summary results are shown in Table 4. We do not include any post-hoc adaptation methods since these results are for the same setting as training.

Our model outperforms PAED. Our model with greedy search outperforms the fine-tuned and zero-shot versions of PAED across all metrics (0.66 triplet accuracy versus 0.61 and 0.06, respectively). PAED in its original zero-shot setting performs poorly. Upon further examination, this is due to the "not" label having the highest likelihood compared to the other labels. The original PersonaExt relation types were detailed (e.g., "school status"), so perhaps the model would perform better with more explicitly named relations (e.g., "attribute of" instead of "characteristic").

Relation-label accuracy is impacted by training prevalence. In the per-relation scores, all models are best at identifying the "Characteristic" and "Routine or Habit" relations, which are the most prevalent in the training dataset. The models are the least accurate at identifying "Experience", "Goal or Plan", and "Not a Relation".

Predicting head entity is a simple task. All models, except PAED (zero-shot) have accuracy higher than 0.95 on identifying the Head entity in the triplet. This is expected as there are only a few options for head entities and the vast majority are "i" or "my" in the training set.

Triplet accuracy is penalized by triplet tail. All models perform poorly on predicting the tail of a triplet, with an accuracy of only 0.55-0.62. This low accuracy lowers the overall triplet accuracy, which is mostly accurate for the head entities and relations, with accuracy above 0.97 and 0.81, respectively. The difficulty in tail predicting is most likely due to the length, which often has several tokes as compared to the 1-3 tokens in the head, and not all the tail entity tokens are taken directly from the context.

D Annotation Guidelines

Annotation guidelines were created by one author after qualitatively evaluating five random characters (sampled with random seed 13). The guidelines are shown in Appendix E. Each triplet was annotated against the provided character name and 3-4 sentence description (i.e., persona profile in Section 5.1). Triplets were annotated individually and not as a whole, e.g., two triplets that are contradictory could still be labeled as reasonable. For example, if the description does not contain information about the character's family, "has sibling sister" and "has no siblings" are both reasonable. Also, triplets were annotated based on the present. If the description specifies "I am a pirate" and the triplet says "I want to be a pirate" it's labeled as a contradiction since the character already is a pirate. Further, some triplets were near-duplicates of each other and were consolidated by merging triplets with first-person heads into one subject "self". Also, triplets were lowercased before merging. Note this consolidation was only performed for annotations and thus the number of triplets differs from those present in Table 15. Table 2 shows example annotations.

E Detailed LIGHT Results

Detailed quantitative intrinsic metrics are in Table 15 and human evaluation results are in Appendix E.

Name	Source of Data	World Setting	N. characters	Dialogue?	Persona?
Dungeons and Dragons (Callison-Burch et al., 2022)	Play-By-Post	Dungeons and Dragons	7168 (est.)	prose	No
LIGHT (?)	LIGHT crowdsource platform	LIGHT	1755	dialogue	structured
TorchLight II (van Stegeren and Theune, 2020)	Torchlight II	Torchlight	82	dialogue	No
Fireball (Zhu et al., 2023a)	Avrae bot on Discord	Dungeons and Dragons	160K	prose	unstructured
KNUDGE (Weir et al., 2023)	The Outer Worlds video game	The Outer Worlds video game	81	dialogue	unstructured
Storium (Akoury et al., 2020)	Storium	Multiple	25,955	prose	unstructured
TVShowGuess (Sang et al., 2022)	TVMegaSite	Multiple		dialogue	No
Star Wars: KOTOR (van Stegeren and Theune, 2020)	Star Wars: KOTOR	Star Wars universe	556	dialogue	No
HLA-Chat (Li et al., 2020)	TV Tropes	Multiple	45,821	dialogue	structured
LiSCU (Brahman et al., 2021)	schmoop, SparkNotes, cliffNotes, LitCharts	Multiple	9,499	prose	no

Table 3: Datasets from the literature considered in this work for persona knowledge graph narrative adaptation. LIGHT stands for Learning in Interactive Games with Humans and Text. KOTOR stands for Knights of the Old Republic. LiSCU stands for Literature Summary and Character Understanding.

		Label				Head	Tail		Triplet
		Precision	Recall	F1	Acc.	Acc.	Overlap	Acc.	Acc.
Our Model	Greedy Search	**0.85**	**0.84**	**0.84**	**0.84**	**0.96**	**0.78**	**0.67**	**0.66**
PAED	(fine-tuned)	0.83	0.83	0.83	0.83	0.95	0.75	0.62	0.61
	(zero-shot)	0.52	0.06	0.03	0.06	0.08	0.07	0.07	0.06

Table 4: Results on the "in-domain" dataset, PersonaExt-PeaCoK. Evaluated on the test split. "Acc": Accuracy. The overall triplet accuracy is the same across decoding methods for our model.

Name	Input	Output
PAED	Context : {context} Head Entity : <mask>, Tail Entity : <mask> , Relation : <mask> .	Head Entity : {head} , Tail Entity : {tail} , Relation : {relation} .
Tokens	[CONTEXT] {context} [HEAD] <mask> [TAIL] <mask> [RELATION] <mask>	[HEAD] {head} [TAIL] {tail} [RELATION] {relation}
Relation-first	[CONTEXT] context [RELATION] <mask> [HEAD] <mask> [TAIL] <mask>	[RELATION] {relation} [HEAD] {head} [TAIL] {tail}
Relation-first-nomask	[CONTEXT] {context}	[RELATION] {relation} [HEAD] {head} [TAIL] {tail}

Table 5: Evaluated input and output prompts for fine-tuning BART for persona extraction task.

Token	Description
[CONTEXT]	context
[RELATION]	relation
[HEAD]	head entity
[TAIL]	tail entity
[characteristic]	character trait
[no_relation]	no relation
[routine_habit]	regularly or consistently do
[goal_plan]	will do or achieve in the future
[experience]	did in the past

Table 6: Tokens added to BART vocabulary for training the persona extraction model. The descriptions for the relation types are from PeaCoK (Gao et al., 2023a). Figure 3 shows an example model input and output.

		Relation				Head	Tail	Triplet
		Prec.	Recall	F1	Acc.	Acc.	Acc.	Acc.
BART	Token prompt	0.81	0.81	0.81	0.81	0.97	0.56	0.55
	Relation-first	0.83	0.83	0.83	0.83	0.98	0.61	0.60
	Relation-first-nomask	0.81	0.81	0.81	0.81	0.98	0.55	0.55
	PAED prompt	0.81	0.81	0.81	0.81	0.97	0.56	0.56
PAED		0.83	0.83	0.83	0.83	0.98	0.62	0.61

Table 7: Summary results for our model (BART) trained with the variety of input and output templates (Table 5) and comparative model PAED trained and tested on the PersonaExt-PeaCoK dataset. Results are on the test set.

Param.	Value
Batch size	32
Epochs	20
Seed	42
Optimizer	AdamW
Learning Rate	$5e-5$
Scheduler	linear
Warmup	1 epoch
weight decay	0
adam_beta1	0.9
adam_beta2	0.999
adam_epsilon	$1e-8$

Table 8: Hyperparameter settings for the persona extraction model. The best model was saved at the end according to evaluation loss on the validation set. Model was trained on 4 GPUs, batch size shown is the effective batch size.

Accepted	Sub-category	Guideline
Yes	Directly	Relates to the persona description
	Reasonable	Character trait given the persona description
No	Contradictory	Candidate persona goes against the persona description. E.g., persona is a dragon but the relation is (I, routine_habit, am a high priestess)
	Unreasonable	Unreasonable trait given character description
	Malformed	Either non-specific, e.g. (self, characteristic, like activity it) or nonsensical relation and tail pairing, e.g. (self, routine_habit, marital status sleeping)

Table 9: Guidelines for the human evaluation of extracted persona information. Evaluators were required to select a sub-category within "Yes" and "No" providing a reason for the selection.

Utterance	PersonaExt Triplet	PeaCoK Triplet
clothes . i am going to auburn for med school .	(i, attend_school, auburn)	(i, routine_habit, attend school auburn)
lol , maybe . i usually wear band shirts and ruffle sleeves , skinny jeans and leggings	(i, favorite, ruffle sleeves)	(i, characteristic, favorite ruffle sleeves)
hmmm . i would travel more if i had the money . you travel or sing ?	(i, want, money)	(i, goal_plan, want money)
i never flew a plane , but i have flow from france to canada where i live	(i, place_origin, france)	(i, experience, place origin france)
do you have friends here , i have lots here . You going back to school ?	(i, misc_attribute, friends)	(i, not, misc attribute friends)

Table 10: Example datapoints from PersonaExt modified to fit PeaCoK format.

Relation	PersonaExt Relations
Characteristic	belief, favorite, favorite_activity, favorite_animal, favorite_book, favorite_color, favorite_drink, favorite_food, favorite_hobby, favorite_movie, favorite_music, favorite_music_artist, favorite_place, favorite_season, favorite_show, favorite_sport, gender, has_ability, has_age, have_chidren, have_children, have_family, have_no, have_no_children, have_no_family, have_no_sibling, have_pet, have_sibling, have_vehicle, health_status, like_activity, like_animal, like_character, like_color, like_drink, like_food, like_general, like_goto, like_movie, like_music, like_read, like_sport, like_sports, like_subject, like_watching, name, physical_attribute, race, scared_of, sexual_orientation, weakness
Experience	has_degree, international_exp, nationality, place_origin, pre_employed_by_company, pre_employed_by_general, previous_profession, raised_by, used_to
Goal plan	want, want_do, want_job, want_no
Not	fall_out, have, industry, misc_attribute, other, work_schedule
Routine habit	allergy_to, attend_school, collect, diet, dislike, dislike_activity, dislike_animal, dislike_color, dislike_drink, dislike_food, dislike_job, dislike_music, dislike_sport, dislike_subject, do_not_do, do_not_drink, do_not_eat, doing, employed_by_company, employed_by_general, get_along, has_hobby, has_profession, job_status, live_in_citystatecountry, live_in_general, live_with, marital_status, member_of, never_do, own, relationship, school_status, teach, worry_about

Table 11: Corresponding PeaCoK relation type for each of the 105 PersonaExt relations. The 'not' category refers to not being mapped to a PeaCoK relation type.

	Train	Dev	Test	Total
Characteristc	0.57	0.57	0.57	0.57
Experience	0.03	0.03	0.03	0.03
Goal or Plan	0.04	0.04	0.04	0.04
Routine or Habit	0.31	0.31	0.31	0.31
Not a Relation	0.04	0.04	0.04	0.04
All	28412	3157	3508	35077

Table 12: Size and label distribution for each split in the PeaCoK-PersonaExt dataset. The Characteristic and Routine or Habit attributes are the most frequent. The train, validation, and test sets for model training were stratified-split according to the labels.

	Top Entity Values
Head	i (2920), my (309), me (90), we (33), my dad (27), my mom (25), mine (17), my mother (17), my parents (16), my brother (8), my father (7), dog (5), parents (4), my wife (4), my boyfriend (4), my family (4), family (3), my husband (2), best friend (2), friends (1), our (1), friend (1), mom (1), my girlfriend (1), us (1), cat (1), my dads (1), my son (1), my grandma (1), my best friend (1)
Tail	have pet dog (39), have pet cat (37), favorite food pizza (24), diet vegan (21), have pet cats (18), like animal dogs (18), marital status married (18), like music country (17), has profession teacher (17), have friends (17), attend school high school (17), like activity hiking (15), has profession nurse (15), like activity video games (15), have vehicle car (14), like activity shopping (14), have pet dogs (14), like animal dog (13), like color blue (13), like activity travel (13), has profession artist (11), have sibling sister (11), physical attribute short (11), like sports basketball (10), like goto beach (10), physical attribute tall (10), like drink coffee (10), like activity reading (10), place origin farm (10)

Table 13: The most prevalent values for the Head and Tail entities in the PeaCoK-PersonaExt dataset. Only the tail entities with at least 10 occurances are shown. There are 30 unique head entitites and 1708 unique tail entities.

		No - Contradictory	No - Malformed	No - Unreasonable	No Agreement	Yes - Directly	Yes - Reasonable	Ratio Accepted
Neutral Removed	Diverse Beam Search	8	38	3	58	9	126	**0.56**
	Greedy Search	7	52	4	78	9	136	0.51
Re-Ranking	Diverse Beam Search	13	68	10	111	11	219	0.53
Baseline	Diverse Beam Search	8	69	11	114	9	217	0.53
	Greedy Search	9	66	9	112	12	198	0.52

Table 14: Qualitative, human evaluations for our model evaluated on the new fantasy narrative setting (LIGHT). Inter-annotator agreement (IAA, Krippendorff's Alpha measured with two annotators) of 0.54. "Ratio Accepted" refers to extracted persona annotated with the "Yes" category (i.e., marked as appropriate by both annotators). An important note is that the numbers are slightly different from Table 1 due to lower agreement on the fine-grained labels versus overall "yes" and "no".

		Per character						Overall			
		Recall	Cov.	Cov. (unique)	1st person	Unique persona	N. persona	Cov.	Cov. (unique)	1st person	N. persona
Neutral Removed	Diverse Beam Search	0.25	0.50	0.46	**0.88**	0.92	37.66	0.49	0.19	**0.88**	22430
	Greedy Search	0.28	0.63	0.58	0.87	0.92	47.61	0.63	0.24	0.87	28476
Re-ranking	Diverse Beam Search	0.33	**0.88**	**0.81**	0.86	0.92	66.80	0.88	0.33	0.86	39680
Baseline	Diverse Beam Search	**0.34**	**0.88**	**0.81**	0.86	0.92	67.09	0.88	0.33	0.86	**40153**
	Greedy Search	**0.34**	0.86	0.79	0.86	0.92	65.53	0.86	0.33	0.85	40068

Table 15: Quantitative results with intrinsic metrics across all models on the new narrative setting (fantasy, LIGHT). "Cov." is short for utterance coverage. The Overall metrics are across all extracted persona and not analyzed for each character, thus the persona recall metric is not applicable.